



Incident report analysis

Instructions

As you continue through this course, you may use this template to record your findings after completing an activity or to take notes on what you've learned about a specific tool or concept. You can also use this chart as a way to practice applying the NIST framework to different situations you encounter.

Summary	The organization experienced a two-hour network outage due to a Denial of Service (DoS) attack, in which a malicious actor exploited an unconfigured firewall to launch a flood of incoming ICMP packets. This attack overwhelmed the internal network, completely disrupting normal traffic and preventing access to network resources. The incident response team quickly contained the threat by blocking incoming ICMP packets, taking non-critical services offline, and prioritizing the restoration of critical network functions. To prevent future incidents, the security team then implemented strict firewall rate-limiting rules, source IP verification, continuous network monitoring, and an IDS/IPS system to filter suspicious traffic.
Identify	Conduct a regular audit of the internal network, including all firewalls, systems, and devices, to identify unconfigured assets (such as the unconfigured firewall in this event) and potential security gaps.
Protect	Implement a new firewall rule to limit the rate of incoming ICMP packets. Additionally, enable source IP address verification on the firewall to check for spoofed IP addresses on incoming traffic.

Detect	Deploy network monitoring software to analyze and detect abnormal traffic patterns. Implement an IDS/IPS system to identify and filter out ICMP traffic based on suspicious characteristics.
Respond	Block incoming ICMP packets to mitigate the active DoS attack. Take non-critical network services offline to reduce network strain, and prioritize restoring critical network services.
Recover	Restore all non-critical network services and affected systems back to normal operation. Continue to monitor the network to ensure normal traffic flow is fully restored and no further issues exist.

Reflections/Notes:

This incident highlighted the critical importance of applying a structured approach, like the NIST Cybersecurity Framework (CSF), to manage and mitigate cyber threats. I learned that simple human errors, such as an unconfigured firewall, can lead to severe network disruptions. By working through the Identify, Protect, Detect, Respond, and Recover functions, I now better understand how to systematically analyze an attack, implement preventative security controls (such as rate limiting and IDS/IPS), and effectively minimize downtime during a security event.